PHASE 3 / HISTORICAL DATASET PREPARATION / 01

Historical dataset preparation

Smart Parking System
Eight-Week Synthetic History

Exactly twelve slots | One-minute observations | Fifteen-minute forecast

Prepared 11 November 2026. This phase provides a reproducible offline historical dataset, causal features, target alignment and chronological evaluation partitions. Every observation is explicitly synthetic. No real-world forecast accuracy or Wokwi runtime success is claimed.

Deliverable / check	Actual result
Raw history	80,640 rows: 14 September through 8 November 2026, Asia/Kolkata (IST).
ML-ready observations	70,515 rows; 37 numeric predictors; target and provenance stored separately.
Final data acceptance	18 critical checks; overall PASS=True.
Software tests	21 tests; passed=True, failures=0, errors=0, skipped=0.
Data / model storage	7,327,764 bytes (7.328 MB); limit 50,000,000 bytes. No model artifact.

The preferred Parquet format could not be used: Windows Application Control blocks the existing PyArrow native DLL. The verified processed artifact is deterministic gzip-compressed CSV. Security policy, dependencies and frozen Phase 1/2 implementation were preserved.

Scope ends at validated ML-ready data. No model training, prediction endpoint, live-data ingestion change, Streamlit integration or notebook generator is included. The existing four notebooks remain title-only.

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Architecture, provenance and memory budget

Stage	Input -> output / responsibility
Seeded simulator	Fixed calendar + seed 42 + twelve-slot state machine -> raw CSV.
Strict raw reader	Timezone normalization and value checks before narrowing to uint8. Invalid input fails rather than being silently repaired.
Causal feature engineering	Current observations + past-only lags/windows + known calendar -> numeric features.
Target and partitioning	Exact t+15 label -> warmup/tail removal -> two purge gaps -> 70/15/15 chronological groups.
Compact storage	One processed CSV.gz plus schema/provenance manifest. Splits are labels, not duplicate files.
Independent validation	Full-artifact arithmetic, transitions, target timestamps, causality interventions, reproducibility and storage checks -> JSON/checklist/PDF.

Separation from real observations

data_source is synthetic in every raw and processed record. The generator has no MQTT or SQLite dependency and does not read, merge, replace or append live Wokwi observations. Live slot status remains the responsibility of the frozen backend. Synthetic data does not repair or conceal the outstanding simulator connectivity issue.

Lightweight representation

A wide minute row stores P1-P12 once, avoiding twelve repeated timestamp/source values per minute. State and counts use uint8; engineered fractional features use float32; repeated labels are categorical on load. Raw CSV and processed gzip CSV are the only full representations. No model, train/validation/test copies, uncompressed processed export or generated EDA images are stored.

Measured DataFrame storage: raw 1,935,550 bytes; processed 7,757,235 bytes. These are table-memory measurements, not a peak-process-RAM benchmark. Temporary arrays used for tests are small and purposeful; reproducible compressed bytes are checked with a hashing sink rather than another dataset file.

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Synthetic arrival and departure methodology

The simulation runs for 56 continuous days. Each minute first removes vehicles whose sampled departure time has arrived, samples Poisson arrival demand, and fills available slots using preference weights from 1.15 (P1) to 0.85 (P12). Excess demand is discarded when capacity is exhausted. Just-departed slots are unavailable until the next minute so all arrivals/departures remain visible in sampled slot transitions.

```
G(h; mu, sigma) = exp(-0.5 * ((h-mu)/sigma)^2)
weekday lambda(h) = 0.0015 + 0.055*G(h;8.8,1.25)
                   + 0.025*G(h;13,1.8) + 0.015*G(h;17.5,1.5)
weekend lambda(h) = 0.001 + 0.025*G(h;12.8,2.3)
                   + 0.015*G(h;18,1.5)
arrivals_attempted(t) ~ Poisson(lambda(h)*daily_demand)
daily_demand ~ LogNormal(-0.5*0.18^2, 0.18)
```

Duration is drawn at arrival: lognormal sigma=0.5, with pre-clipping mean 300 minutes for weekday arrivals from 06:00 to before 11:00; 110 minutes for other weekday arrivals; 100 minutes on weekends. Values are rounded and clipped to 10-720 minutes. The latent scheduled departure is never exported as a feature. Slots start vacant at midnight and are never reset at day boundaries.

Measured synthetic behavior	Value
Weekday business hours 09:00-16:59	6.4073 occupied slots on average
Weekend same hours	1.7805 occupied slots on average
Weekday night 00:00-04:59	0.2166 occupied slots on average
Lag-one autocorrelation	0.998602
Completed stays / dwell	1042; min 21, median 141.0, max 720 minutes

These checks demonstrate the intended synthetic patterns and persistence only. They are not empirical evidence that a particular real site follows these distributions. Boundary-censored stays are excluded from completed-dwell summaries, not discarded from state history.

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Raw and processed schema

Raw column(s)	Meaning / type
timestamp	One local minute with explicit +0530 UTC offset; timezone-aware on load.
data_source	Constant synthetic; categorical on load.
P1-P12	Twelve uint8 values: 0 VACANT, 1 OCCUPIED. All twelve exist on every row.
total_occupied	uint8 sum of the twelve states; always 0-12.
arrivals / departures	uint8 counts of 0->1 and 1->0 transitions in that minute; each 0-12.

Processed table: 42 columns

The processed table has timestamp and data_source, 37 allowlisted predictors, target_timestamp, occupied_15min_ahead and split. Target is uint8, range 0-12. Split is train, validation or test. Detailed per-column dtypes are recorded in phase3_dataset_manifest.json. Metadata and targets must not be passed to a model merely because a column is numeric.

Cleaning and validation policy

Generated raw observations contain no missing values. Input validation rejects missing/nonbinary slots, invalid count ranges, duplicate or irregular timestamps and non-synthetic provenance. Numeric ranges are checked before uint8 conversion, preventing an invalid value such as 256 from wrapping into an apparently valid zero. Time is normalized to Asia/Kolkata; no DST discontinuities occur in this fixed interval.

No backward fill, interpolation or future-based imputation is performed. Initial lag/rolling gaps are expected and removed as a declared warmup. The final 15 origins have no future label and are excluded from ML-ready rows. Boundary purge rows remain in raw history and may supply causal past context.

Storage format decision

Raw is ordinary UTF-8 CSV. Processed is CSV.gz with fixed gzip header mtime=0, empty embedded filename and stable numeric formatting. Nine significant digits preserve float32 values on round-trip. The loader restores dtypes/timezones from the manifest. A Zstandard Parquet code path exists, but Windows blocks PyArrow and that path is not runtime-verified here. No dependency installation or security bypass was attempted.

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Feature calculations and leakage prevention

Predictor group	Calculation / information available at forecast origin t
Current state / counts: 15	P1-P12, their total, arrivals and departures observed at t.
Calendar: 8	Hour, minute, weekday (Mon=0), weekend; daily sine/cosine using minute-of-day / 1440; weekday sine/cosine using weekday / 7.
Lag: 7	total(t-k) for k=1,5,15,30,60,1440,10080 minutes.
Trailing mean: 4	Mean of total(t-w) through total(t-1), w=5,15,60,1440.
Trailing std: 2	Population standard deviation of the same past-only window, w=15,60; ddof=0.
Change: 1	total(t)-total(t-15), with signed/floating arithmetic.

```
mean_w(t) = sum(total(i), i=t-w,...,t-1) / w
std_w(t) = sqrt(mean(total(i)^2) - mean_w(t)^2)
label(t) = total(t+15 minutes)
label_time(t) = timestamp(t+15 minutes)
```

All predictor values use observations at or before t or calendar facts known at t. For example, a five-minute mean at 09:00 uses 08:55-08:59. A weekly lag uses the same minute seven days earlier. Target_timestamp, target, split, source and raw timestamp are outside FEATURE_COLUMNS. Scheduled departure times and future simulation parameters are not predictors.

There are no fitted scalers, global occupancy aggregates, imputers, random splits or centered rolling windows. Any preprocessing added in a later model phase must be fitted on training data only. Availability of the current t observation is an explicit forecast-origin assumption; changing that deployment timing would require shifted features.

Leakage tests actually executed

Every stored lag is checked against exact raw indices. All rolling values are independently checked using cumulative sums/squared sums, not just the feature function. Every calendar/current/change feature is verified. Future observations are altered while every past feature is compared; prefix-only feature computation must equal the same prefix computed with later rows present. Target alignment is checked for every processed row.

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Chronological 70/15/15 split and purges

```
Raw observations                     80,640
Remove weekly-lag warmup            -10,080
Remove targetless final 15 minutes      -15
Reserve two 15-minute boundary gaps     -30
Final usable forecast origins        70,515
Train = floor(0.70 * 70,515)          49,360
Validation = floor(0.15 * 70,515)     10,577
Test = remainder                     10,578
```

Split	Rows / fraction	First / last origin (IST)	Last label time (IST)
train	49,360 / 69.99929%	2026-09-21 00:00:00 / 2026-10-25 06:39:00	2026-10-25 06:54:00
validation	10,577 / 14.99965%	2026-10-25 06:55:00 / 2026-11-01 15:11:00	2026-11-01 15:26:00
test	10,578 / 15.00106%	2026-11-01 15:27:00 / 2026-11-08 23:44:00	2026-11-08 23:59:00

Integer rounding accounts for the tiny deviation from 70%, 15%, 15%. There is no shuffling. Origins 25 October 06:40-06:54 and 1 November 15:12-15:26 are excluded. The last origin before each gap and first origin after it are 16 minutes apart.

The final training label is 25 October 06:54, strictly before validation starts at 06:55. The final validation label is 1 November 15:26, strictly before testing starts at 15:27. Thus neither preceding partition has a label whose outcome occurs at or after the next partition starts.

Validation/test lags may read earlier observed history from the previous partition or purge gap. This is causal context available at the later forecast origin, not leakage. No fitted transform sees held-out rows, and no model has yet been trained or tuned. The test period must remain untouched during future model selection.

Raw history still represents all eight weeks. The first seven days are retained as lookback context, even though no ML-ready origin is emitted during that warmup. Last raw timestamp is 8 November 23:59; last processed origin is 23:44, whose exact label is 23:59.

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Full-artifact acceptance checklist

Critical verification	Result	Evidence scope
eight weeks one minute	PASS	All 80,640 raw timestamps
all twelve slots	PASS	P1-P12 columns and binary raw states
synthetic provenance	PASS	Both complete tables
missing values and numeric ranges	PASS	All values, all predictor columns
total equals slot sum	PASS	Every raw minute
slot arrivals departures and dwell	PASS	Every slot transition and completed stay
synthetic daily weekly patterns	PASS	Business/weekend/night means and persistence
exact fifteen minute target	PASS	All 70,515 future timestamps and labels
current features match observation	PASS	Every current slot/count feature
all lags are past only	PASS	All seven lags at every origin
all rolling windows exclude future	PASS	Independent cumulative-sum oracle
calendar and change features	PASS	All known-calendar and delta values
future intervention and prefix invariance	PASS	All 37 features; 10,181 prefix rows
chronological 70 15 15 purged boundaries	PASS	Counts, ordered origins and separated labels
schema predictor allowlist	PASS	Exact schema and excluded metadata/targets
fixed seed full reproducibility	PASS	Regenerate all 80,640 observations
artifact checksums	PASS	Raw/processed SHA-256 match manifest
storage below 50 MB	PASS	Data + model directories below 50 MB

The machine-readable record is reports/evidence/phase3_dataset_validation.json; the human checklist is reports/phase3_acceptance_checklist.md. A failed critical check makes validation exit nonzero. Software execution, frozen-scope preservation and final document review are recorded separately in phase3_tests.json and phase3_final_checks.json.

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Software tests and slot-level transition evidence

Suite	Tests	Coverage
FeatureTests	9	Each lag/window, causal prefix/future intervention, exact labels, partitions and invalid input
GenerationTests	7	Calendar, twelve states, totals, transitions/dwell, seed behavior and provenance
SerializationTests	5	CSV round-trip, overflow rejection, no overwrite, stored hashes and deterministic processed bytes

Actual archived run: 21 tests in 13.156 seconds; passed=True; 0 failures, 0 errors, 0 skips. Test IDs, runner output and source hashes are saved, rather than replaced by illustrative success text.

Slot	Arrivals	Departures	Occupied fraction
P1	100	100	0.22428
P2	93	93	0.20857
P3	91	91	0.20362
P4	95	95	0.18994
P5	84	84	0.17650
P6	84	84	0.17502
P7	86	86	0.20156
P8	89	89	0.21022
P9	82	82	0.19587
P10	82	82	0.19942
P11	79	79	0.18905
P12	77	77	0.17774

The full-seed regeneration content hash matches raw input and manifest. Recomputing all processed features and writing to a hashing sink yields the same gzip byte hash and size as the stored artifact, without a duplicate output file.

Initial Parquet preparation failed because Application Control blocked PyArrow; the empty new partial output was removed. The initial sandbox test attempt encountered temporary-directory access/cleanup restrictions. The permitted rerun passed all tests. Security controls and installed dependencies were not changed.

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Measured outputs and reproducibility commands

Data/model file	Actual bytes
phase3_dataset_manifest.json	4489
synthetic_parking_8weeks_ml.csv.gz	2077563
synthetic_parking_8weeks_raw.csv	5245712

Total = 7,327,764 bytes = 7.327764 MB = 6.988 MiB. Limit is decimal 50 MB. No model files were created. Reports/test evidence are separate documentation artifacts, not duplicate datasets.

Normal commands from repository root

```
.\.venv\Scripts\python.exe -B -m ml.generate_data
.\.venv\Scripts\python.exe -B -m ml.prepare_data
.\.venv\Scripts\python.exe -B -m ml.validate_data
.\.venv\Scripts\python.exe -B -m unittest discover -s ml/tests -v
```

Commands refuse existing outputs. To reproduce without replacing the released artifacts, choose new paths:

```
.\.venv\Scripts\python.exe -B -m ml.generate_data `
  --output tmp/repro_raw.csv
.\.venv\Scripts\python.exe -B -m ml.prepare_data `
  --raw tmp/repro_raw.csv --output tmp/repro_ml.csv.gz `
  --metadata tmp/repro_manifest.json
.\.venv\Scripts\python.exe -B -m ml.validate_data `
  --manifest tmp/repro_manifest.json --evidence tmp/repro_validation.json `
  --checklist tmp/repro_checklist.md
```

Content fingerprints and environment

Raw SHA-256
eb2b30880602b263d8025396f9def658273b7d996129a77c6f4bef3a78c3aab4

Processed SHA-256
882274bf29e7fc026e4b6aa91e40de6ebdd738aa61564f59af511484274881d6

Recorded versions: numpy 2.5.4, pandas 3.1.1, pyarrow 26.0.1. Seed 42; fixed start date and timezone. Reproduce with these versions; generator changes can intentionally change data.

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Assumptions, limitations and continuation

Assumption / limit	Practical implication
Synthetic, uncalibrated demand	Temporal plausibility is checked; real-site demand, generalization and forecast accuracy are not established.
Perfect binary observation	No UNKNOWN states, sensor dropout, delayed MQTT, gateway failures or distance noise. Real-data preprocessing requires an explicit missingness/freshness policy.
Fixed calendar and demand families	No holidays, weather, exceptional events, tariffs, reservations, special access or structural demand shifts.
Capacity and arrivals	Twelve spaces only; excess demand is lost. No queue, user identifiers or travel behavior model.
Current minute available	Feature timing assumes the t observation has arrived before issuing its t+15 forecast.
One chronological holdout	No model training, hyperparameter search, cross-validation or real-world performance claim.
Parquet policy restriction	Only gzip CSV was verified. Optional Parquet needs a permitted native Arrow environment; no bypass is proposed.

Acceptance and preservation

Data acceptance requires every listed critical check to pass. Final phase acceptance additionally requires successful software tests, unchanged frozen Phase 1/2 files, preservation of live observations, a visually inspected PDF and the storage budget. Final verification evidence and the checklist record those outcomes before the one-file commits and single push.

Frozen ESP32/firmware/circuits, backend, database configuration, dependency snapshot, Phase 1/2 PDFs/evidence and existing notebooks are not edited. Only ML data-preparation implementation and new Phase 3 documentation/evidence are added; the shared reports index gains a Phase 3 reference.

Next phase

Use FEATURE_COLUMNS as X and occupied_15min_ahead as y; select rows by split. Fit any learned preprocessing on training only, tune on validation, and evaluate the untouched test period once. Compare with persistence and seasonal baselines. Establish real-site validation before any operational accuracy or live forecast claim. No backend or Streamlit integration is part of this deliverable.

Reference files: ml/README.md; ml/generate_data.py; ml/features.py; ml/prepare_data.py; ml/validate_data.py; ml/tests/test_phase3.py; manifest; validation/test/final-check JSON files; acceptance checklist. This report is generated by reports/generate_phase3_report.py using the bundled ReportLab environment, leaving project dependencies unchanged.